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Key Points:

- A new method evaluates the significance of earthquake seasonality adopting uncertainty derived from a stochastic declustering scheme
- The background seismicity rate of San-in district in Japan shows significant semiannual variation that increases in spring and autumn
- Seasonal variations in seismicity can be explained by heavy precipitation in autumn and snow melting in spring

Supporting Information:

- Supporting Information S1

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Seasonal Variations in Crustal Seismicity in San-in District, Southwest Japan

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Abstract A growing body of evidence suggests that seismicity is seasonally modulated in a variety of tectonic environments. Identifying cyclic variations in seismicity improves our understanding of the physics of earthquake triggering. We explored seasonal modulation of crustal seismicity in San-in district, southwest Japan, using a new method that adopts uncertainties derived from a probability-based declustering procedure. We determined that semiannual variation in background seismicity rate, which increases in spring and autumn, is statistically significant from 1980 to 2017. The frequency distribution of large historic and modern earthquakes shows a similar pattern to recent background seismicity, suggesting that seismicity in San-in district has shown seasonal variations for over 150 years. These observations can be explained by increasing pore pressure within fault zones, caused by precipitation in autumn and decreasing surface mass due to snow melting in spring.

Plain Language Summary Seismic activity is seasonally modulated by surface processes. To explore earthquake seasonality, it is necessary to choose only independent earthquakes from entire seismicity catalog; this selection process is called earthquake declustering. Here, we introduce a new statistical declustering method to evaluate the significance of seasonal modulation of shallow seismicity, which includes an estimate of the uncertainty associated with earthquake declustering. We apply this technique to shallow crustal seismicity in San-in district, southwest Japan. The rate of independent earthquakes from 1980 to 2017 shows significant semiannual variation, with increases in spring and autumn. The distribution of large historic and modern earthquakes shows similar variations, suggesting that seismicity in San-in district has been related to the seasons for at least 150 years. This seasonality can be attributed to heavy precipitation in autumn and snow melting in spring. Seasonal modulation of small earthquakes should be monitored to investigate temporal changes in the probabilities of large earthquakes in San-in district.

1. Introduction

It has been reported that surface processes, such as ice and snow load, heavy rainfall load, atmospheric pressure, and surface temperature, seasonally modulate seismic activity in the crust (e.g., Amos et al., 2014; Ben-Zion & Allam, 2013; Bettinelli et al., 2008; Gao et al., 2000; Heki, 2003; Johnson et al., 2017a). Cyclic variations in seismicity could provide insight into the physics of earthquake triggering.

The clustering of earthquakes in space and time can mask or introduce a bias to evaluation of seasonal variations caused by long-term or subtle changes in stress or fault strength. Therefore, declustering of seismicity is required for accurate evaluation of seasonal variations in seismicity. Various declustering schemes have been proposed based on the spatial and temporal distances between events (e.g., Gardner & Knopoff, 1974; Marsan & Lengliné, 2008; Ogata, 1988; Reasenberg, 1985; Zaliapin & Ben-Zion, 2013). However, these declustering methods produce different sets of declustered catalogs, resulting in nonunique estimates of temporal variations in background seismicity (van Stiphout et al., 2011; Zhuang et al., 2002). Uncertainties derived from a declustering scheme should be incorporated into the evaluation of seasonal modulation of seismicity, but this has been rarely considered in previous studies. In the present study, we use a probability-based declustering procedure based on the work of Zhuang et al. (2002) to evaluate the significance of the seasonal variations, adopting uncertainties derived from the declustering scheme.

San-in district (Figure 1), southwest Japan, is an active tectonic zone characterized by a large shear strain rate of 1.0×10^{-7} /year estimated from geodetic measurements (Nishimura & Takada, 2017), as well as high seismicity rate showing clear lineaments along the coast line (Kawanishi et al., 2009; Ogata et al., 2003). This active zone has recently hosted several large crustal earthquakes, including the 2000 Western Tottori earthquake (JMA (the Japan Meteorological Agency) magnitude M_j 7.3) and the 2016 Central Tottori earthquake

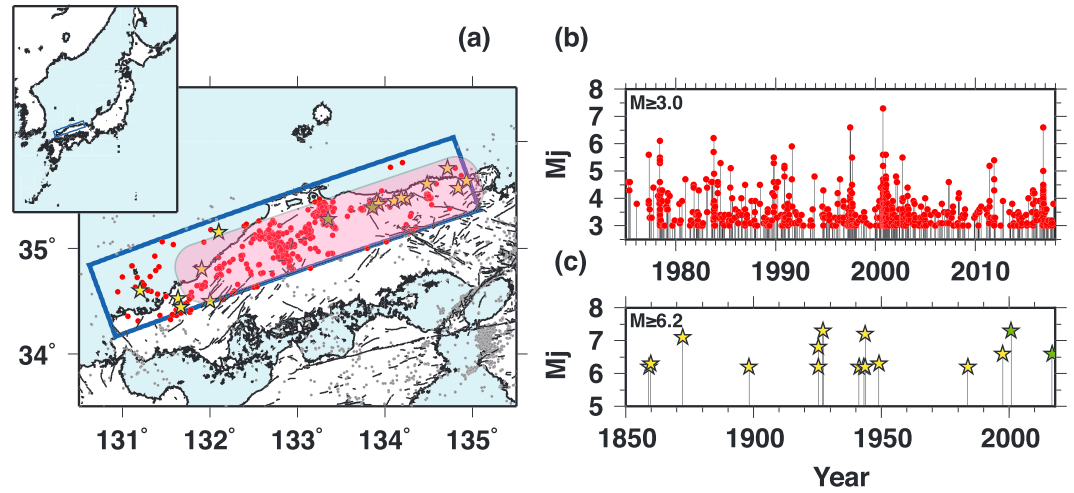


Figure 1. (a) Seismotectonic setting of San-in district, southwest Japan. The blue rectangle outlines the study area. Red and gray dots are epicenters from the earthquake catalog ($M_j \geq 3.0$) determined by the Japan Meteorological Agency (JMA) from January 1980 to December 2017. Yellow stars denote epicenters of large historic earthquakes ($M_j \geq 6.2$) from January 1850 to December 2017, based on the earthquake catalogs of Chronological Scientific Tables and the JMA. Pink shadow indicates the San-in shear zone detected by geodetic measurements (Nishimura & Takada, 2017). Major active faults are indicated by black lines. Inset shows the regional location of the study area (blue square). (b) Magnitude-time plot of earthquakes represented by red dots in Figure 1a. (c) Magnitude-time plot of earthquakes represented by yellow stars in Figure 1a.

(M_j 6.6). Previous studies (Matsumura, 1986; Ogata, 1983) reported seasonal variations in seismicity in the Inner Zone of southwest Japan using a model that includes a seasonal component in the background rate as a function of time, suggesting a weak correlation between background seismicity rate and precipitation. In this work, we reexamine seasonal variations in crustal seismicity in San-in district using a well-determined earthquake catalog and discuss the physical mechanisms of earthquake triggering in the region.

2. Methods

We first fit the observed seismicity rate by adopting a space–time Epidemic Type Aftershock Sequence (ETAS) model (e.g., Ogata, 1998; Zhuang et al., 2004). This model is a point-process model that includes the Omori-Utsu law (Utsu, 1961; Utsu et al., 1995) and a branching process in which each earthquake has the ability to generate its own aftershocks.

The rate of earthquake occurrence at time t is expressed by

$$\lambda(t, x, y | H_t) = \mu(x, y) + \sum_{t_i < t} \kappa(M_i) g(t - t_i) f_{ds}(x - x_i, y - y_i; M_i) \quad (1)$$

$$\kappa(M) = A e^{\alpha(M - M_c)} \quad (2)$$

$$g(t) = \frac{p-1}{c} \left(1 + \frac{t}{c}\right)^{-p} \quad (3)$$

$$f_{ds}(x, y; M) = \frac{q-1}{\pi D^2 e^{\gamma(M - M_c)}} \left(1 + \frac{x^2 + y^2}{D^2 e^{\gamma(M - M_c)}}\right)^{-q} \quad (4)$$

where $\mu(x, y)$ is the background intensity, which is a function of space but not time. The second term in (1) expresses the rate of aftershock occurrence. $\kappa(M)$ defines the expected number of aftershocks; $g(t)$ is the probability density function form of the Omori-Utsu law, which expresses the occurrence times of aftershocks; $f_{ds}(x, y; M)$ is the location distribution of the aftershocks; and H_t is the history of occurrence times up to time t and their corresponding locations and magnitudes. The seven parameters A , c , α , p , D , q , and γ are determined by Maximum Likelihood Estimation (MLE) with a hypocenter catalog. The log-likelihood is given by

$$\log L = \sum_k \log \lambda(t_k, x_k, y_k) - \int_0^T \iint_S \lambda(t', x, y) dx dy dt' \quad (5)$$

where the summation index k includes all events in study region S in time window $[0, T]$. We use an iterative

algorithm that simultaneously estimates the background rate and the branching structure. The details are described in Zhuang et al. (2002).

For the rate of earthquake occurrence fitted by our ETAS model (the modeled seismicity), we applied probability-based declustering algorithms following Zhuang et al. (2002): for each earthquake $i = 1, 2, \dots, N$, we calculate φ_i , the likelihood that each event is a background event, which is given explicitly by

$$\varphi_i = \frac{\mu(\mathbf{x}_i, \mathbf{y}_i)}{\lambda(\mathbf{t}_i, \mathbf{x}_i, \mathbf{y}_i | \mathbf{H}_{t_i})} \quad (6)$$

where t_i is the occurrence time of event i . We generate N uniform random probabilities $U_1, U_2, \dots, U_N$ in the interval $[0, 1]$. If $U_i < \varphi_i$, we keep the i th event as a background event; otherwise, we delete it from the catalog as an aftershock.

A stochastic declustered catalog constructed by the above procedure is nonunique because it depends on the random numbers used to select events. However, we can easily produce many declustered catalogs by repeating the random declustering. From these catalogs, we can evaluate the statistical significance of a specific property included in the declustered catalog. In this study, we propose a new method to evaluate the statistical significance of seasonal variations in background seismicity rate. The algorithm is as follows. (1) Calculate the background rate in each month for each declustered catalog. (2) Perform a Fourier transform on the time series of monthly background rates. (3) Construct the probability density distribution of the amplitudes of the cosine and sine functions for a particular period (e.g., the annual period). (4) Fit a Gaussian distribution to the probability density distribution and evaluate the significance of seasonal variations.

As an earthquake catalog, we use the hypocenter catalog compiled by JMA. We apply the space-time ETAS model to earthquakes within the blue rectangular box in Figure 1a with $M_j \geq 3.0$ and depths ≤ 20 km during 1980–2017 (the number of events is 976). Earthquakes immediately outside of this region (130.5 – 135.5°E , 33.5 – 36.5°N) that occurred during 1975–2017 are used as the precursory occurrence history of the space-time ETAS model. The estimated parameters A , c , α , p , D , q , and γ are 4.34×10^{-1} events, 2.53×10^{-3} days, 1.08 M^{-1} , 1.12 , 3.03×10^{-3} degrees, 1.82 , and $8.39 \times 10^{-1} \text{ M}^{-1}$, respectively. Using the above parameters, we create 10,000 stochastic declustered catalogs and average the time series using the algorithm described above (Figure 2a).

From Figure 2a, the background rate behaves almost homogeneously in time, indicating that intensive aftershocks following major earthquakes (e.g., large earthquakes in 2000 and 2016) are removed from the original catalog. Note that most of the background seismicity is located onshore (Figure S1 in the supporting information). Figure 2b shows the histogram of background rates averaged over the interval 1980–2017 and binned by month. To evaluate the significance of the seasonal variations, we take the Fourier transform of the time series of each declustered catalog and construct a probability density distribution of the coefficients of the cosine and sine functions of the period we focus on. Figure 2c shows the probability density distributions of the semiannual variation coefficients calculated from background seismicity rate time series for 1980–2017. In Figure 2c, the distance from the origin corresponds to the amplitude of the semiannual period, and the angle measured counterclockwise from the positive horizontal axis is the time lag of the background rate measured from mid-January (or mid-July). If the distribution is near the origin, it follows that there is no significant seasonal variation, because there is a high probability that the amplitude of the semiannual period is nearly zero. Therefore, we should quantitatively evaluate the distance between the centroid of the distribution and the origin (hereinafter called δ) to define the significance of the variations. We fit a two-dimensional Gaussian to the probability density distribution, given by

$$f(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-\mu_x)^2 + (y-\mu_y)^2}{2\sigma^2}\right) \quad (7)$$

where x and y are the cosine and sine coefficients of a particular period (e.g., semiannual period), σ is the standard deviation, and (μ_x, μ_y) is the centroid. We define that δ larger than 3σ is significant because the 3σ confidence interval contains $\sim 99\%$ of the values. We focus on the significance of the annual and

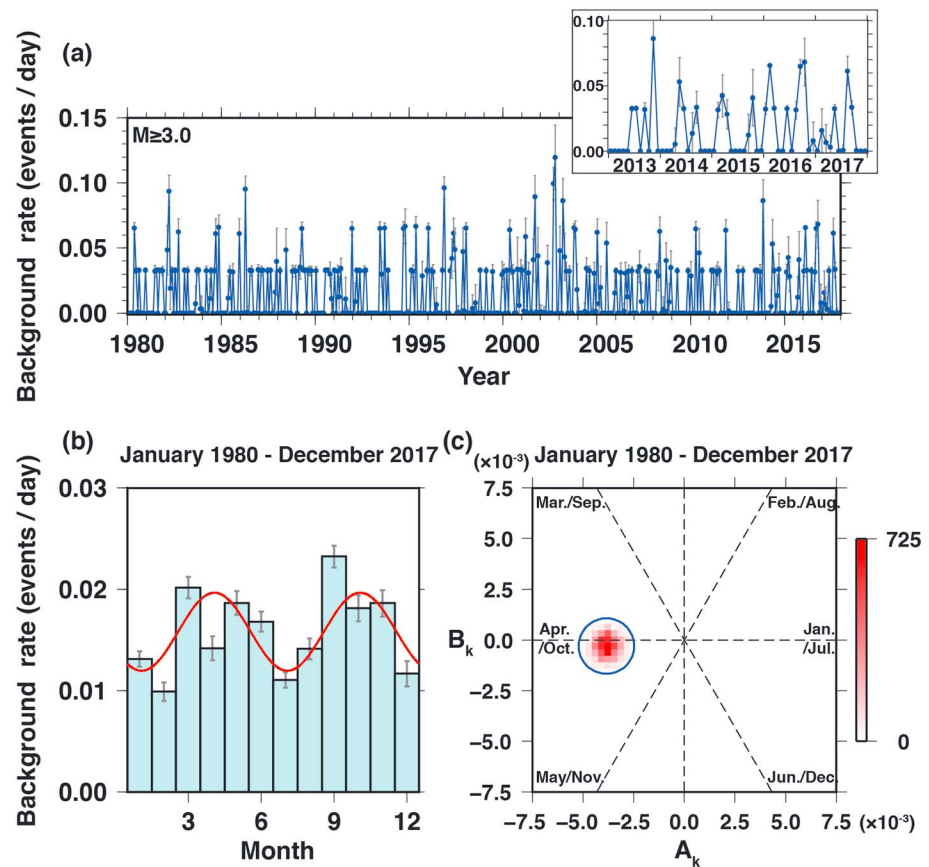


Figure 2. (a) Time series of background seismicity rate for $M_j \geq 3.0$ earthquakes in San-in district, Japan (blue rectangle in Figure 1a). The gray vertical bars around each point show the 1σ uncertainties in the background seismicity rate. The inset shows an enlarged view of the background rate time series from 2013 to 2017. (b) Frequency distribution of monthly background seismicity, averaged over the period January 1980 to December 2017. Red curve denotes the summation of the offset and semiannual variations calculated from background seismicity time series in (a). (c) Probability density distribution of the semiannual modulation coefficients of a cosine function (A_k) and sine function (B_k) calculated from 10,000 declustered catalogs during the period January 1980 to December 2017. The color bar shows the number of coefficients in each bin. The blue contour contains almost 99% of the values (3σ confidence interval). The angle measured counterclockwise from the positive x axis is the delay from mid-January (or mid-July).

semiannual variations. To verify the stability of the variations, we perform the same analysis using data divided into two different time periods: 1980–1997 and 1998–2017.

3. Results

In Figure 2c, δ is 8.5σ , much greater than the 3σ significance criterion. Thus, we conclude that the background rate during this period shows significant semiannual variations. From the frequency distribution of the monthly background rate (Figure 2b), we can see that the background rate increases in spring and autumn. The amplitude of the semiannual period is $\sim 24\%$ of the mean background rate (red line in Figure 2b). Moreover, the subplots in Figure 3 show the frequency distributions of monthly background seismicity rates during 1980–1997 and 1998–2017, indicating that during both periods the background rate also increases in spring and autumn, as for the entire catalog. Figure S2 demonstrates that δ is 6.7σ during 1980–1997 and 6.6σ during 1998–2017, suggesting that the background rate in San-in district has semiannual variation that is stable on decadal timescales. The existence of semiannual variation can also be recognized directly from the inset in Figure 2a, which shows a time series of background rates during the period 2013–2017. For the annual variation, δ is 2.5σ , indicating that annual variation in the background rate is statistically insignificant during the study period (Figure S3).

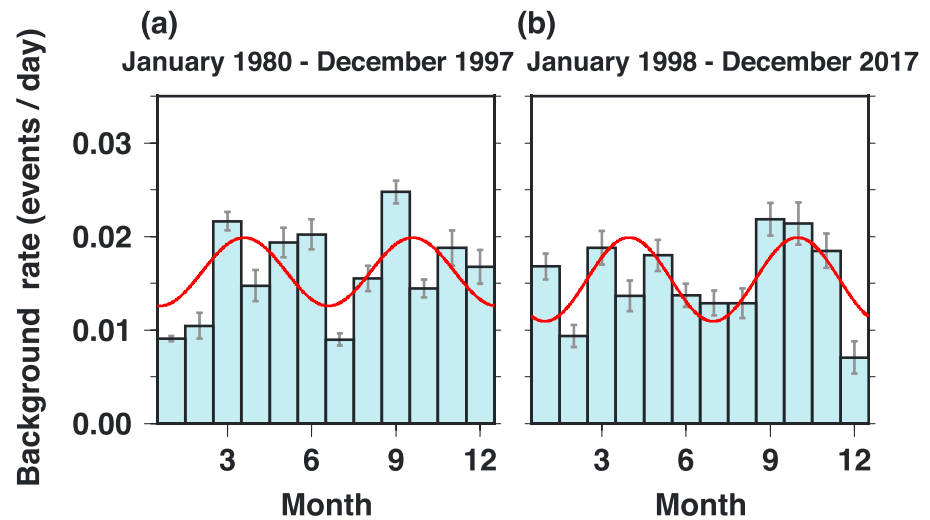


Figure 3. Frequency distribution of monthly background seismicity rate averaged during the two different periods (a) January 1980 to December 1997, and (b) January 1998 to December 2017. The red curve in each panel denotes the summation of the offset and semiannual variations calculated from the time series of background seismicity rates (Figure 2a) for each period. Gray bars have the same meanings as in Figure 2a.

4. Discussion

We now verify the validity of the proposed method by applying it to seismicity in the Sierra Nevada-Eastern California Shear Zone (SN-ECSZ), where annual and semiannual variations in seismicity were previously reported using Multifrequency Periodogram Analysis (MFPA; Dutilleul et al., 2015) during 1990–2001. We use hypocenters compiled by the Advanced National Seismic System (ANSS) Composite Catalog. We apply a space–time ETAS model to seismicity in the same polygonal study area as used by Dutilleul et al. (2015) for $M \geq 2.5$ earthquakes during 1980–2010. Earthquakes peripheral to the target region (116° – 124° W, 34° – 40° N) are used as the precursory occurrence history of the space–time ETAS model. The estimated parameters A , c , α , p , D , q , and γ are 4.93×10^{-1} events, 5.94×10^{-3} days, 1.20 M^{-1} , 1.10, 2.10×10^{-3} degrees, 1.62, and 1.13 M^{-1} , respectively. Figure 4a shows the time series of background seismicity rate averaged over 10,000 stochastic catalogs. The rate shows an increase around summer 1992, corresponding to intense seismicity dynamically triggered by the 1992 M_w 7.3 Landers earthquake (Hill et al., 1993). The frequency distribution of the background rate averaged over the interval 1990–2001 shows two amplitude peaks in summer and winter (Figure 4b). We evaluate the significance of the annual and semiannual variations in background rate during 1990–2001 by applying our method. Based on δ of the annual and semiannual variation coefficients, the background rate in SN-ECSZ has a significant semiannual variation (4.6σ ; Figure 4c), whereas there is relatively weak annual variation in the background rate (2.5σ ; Figure S4). The amplitude of the semiannual period is $\sim 12\%$ of the mean background rate (red line in Figure 4b). Our demonstration of a significant semiannual variation is consistent with the results of Dutilleul et al. (2015), which confirms that the present method is suitable for evaluation of seasonal periodicity. Our method finds weaker annual variation in seismicity than semiannual one, whereas Dutilleul et al. (2015) suggested that annual variation is predominant. These differences between the studies could be attributed to the treatment of the uncertainty associated with the selection of background seismicity (i.e., different choices of declustering methods).

San-in district, where the background seismicity rate increases in spring and autumn, has experienced many large earthquakes in the past (e.g., the 2000 M_j 7.3 Western Tottori earthquake). Figure 5 shows frequency distribution of large historic and modern earthquakes ($M_j \geq 6.2$) during 1850–2017 in San-in district, compiled from the Chronological Scientific Tables (National Astronomical Observatory, 2016); Figure 1c shows the corresponding M – t diagram. The frequency distribution has a similar pattern to the recent background seismicity rate (Figure 2b). The trend of the frequency distribution does not change if we use large earthquakes from the JMA catalog (e.g., $M_j \geq 6.2$ during 1923–2017; see Figure S5). Although there are too few large earthquakes to evaluate statistical significance, the visual similarities between Figures 2b and 5

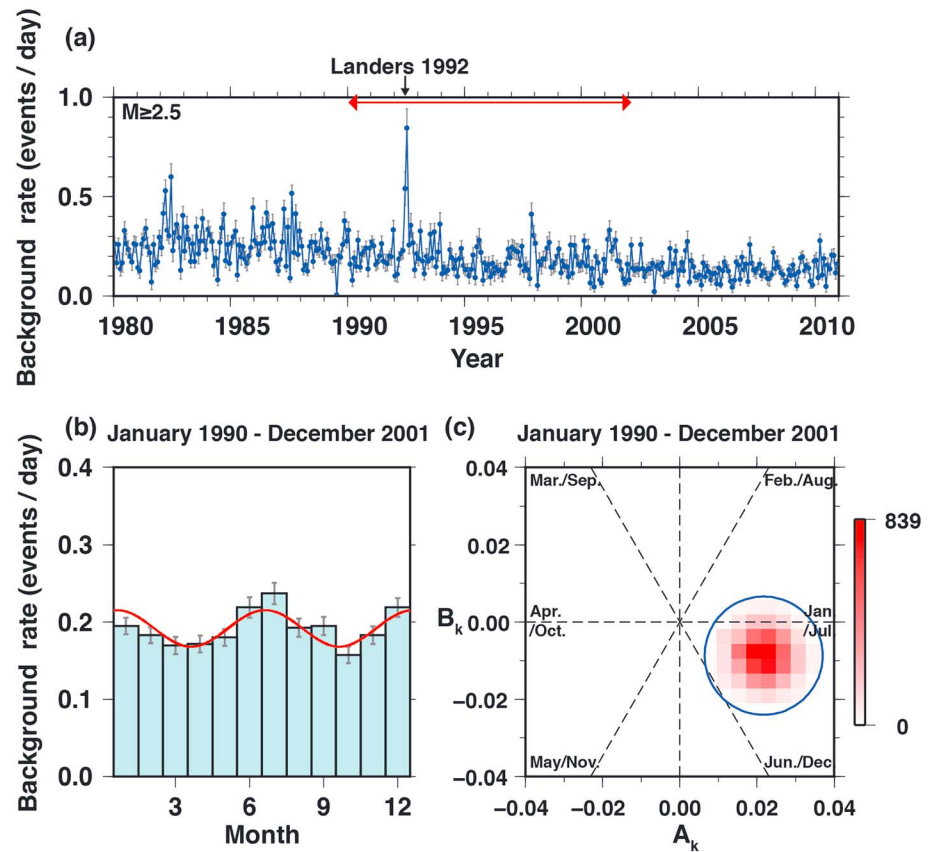


Figure 4. (a) Background seismicity rate of $M \geq 2.5$ events in the Sierra Nevada–eastern California Shear Zone (same study area as considered by Figure 1 of Dutilleul et al., 2015). The gray vertical bars extending from each point give the 1σ uncertainties in the background seismicity rate. The red arrow shows the time period when we analyzed the seasonal variation. (b) Frequency distribution of monthly background seismicity rate from January 1990 to December 2001. Red curve denotes summation of offsets and semiannual variations calculated from time series of background seismicity rate in (a) for the same period. (c) Probability density distribution of semiannual modulation coefficients of a cosine function (A_k) and sine function (B_k) calculated from 10,000 declustered stochastic catalogs during the period January 1990 to December 2001. The color bar shows the number of coefficients in each bin. Blue contour contains almost 99% of the values (3σ confidence interval). The angle measured counterclockwise from the positive x axis corresponds to the time delay measured from mid-January or mid-July.

imply that the seismicity rate in San-in district has shown seasonal variations for >150 years. These features well match with Johnson et al. (2017a) suggesting that both recent seismicity ($M \geq 2.0$) and historic seismicity ($M \geq 5.5$) in California are seasonally modulated.

The variations revealed by the present study are consistent with the results of Ogata (1983), showing that the background seismicity rate of the Inner Zone of Southwest Japan exhibits seasonal changes, to some degree. Ogata (1983) proposed that these variations correlated with increased precipitation. The infiltration of rainfall into the crust or crustal cracks could increase pore pressures within fault zones, leading to weakened fault strength and increased seismicity; indeed, Hainzl et al. (2006, 2013) demonstrated that seismicity below Mount Hochstaufen, Southeast Germany, correlates with the calculated spatiotemporal diffusion of pore pressure after heavy precipitation.

In San-in district, precipitation increases mainly in July and September, due to the rainy and the typhoon seasons, respectively; these are both earlier than the autumnal peak in the background seismicity rate. A time lag of a few months can be explained by the migration of precipitation from the surface to the seismogenic zone, at depths around 8 km (Figure S6). Hainzl et al. (2006) reported that the observed numbers of daily earthquakes are correlated with daily rainfall at a time lag of +8 days. Considering a theoretical prediction for a distance r of the propagating front of pore pressure perturbation from a source, that is, a step increase in rainfall at the surface, the time lag required for fluid diffusion can be derived as $r^2/4\pi\eta$

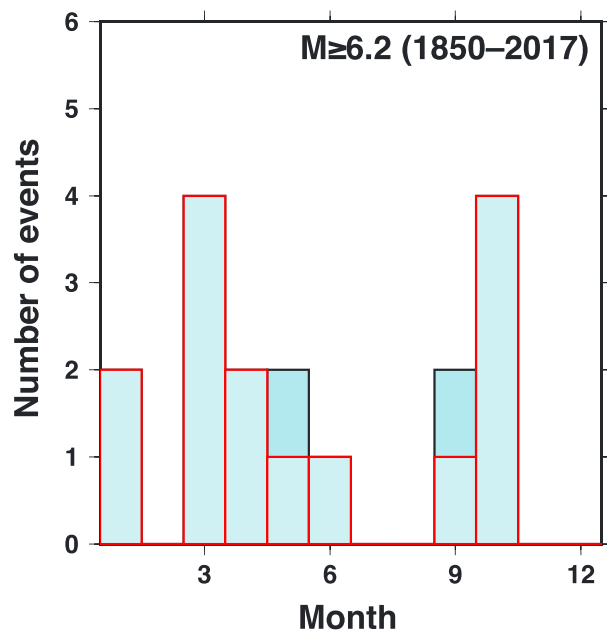


Figure 5. Frequency distribution of large historic earthquakes ($M_j \geq 6.2$) versus month in San-in district, Japan. Red bars exclude potential aftershocks from the original histogram.

(Shapiro et al., 1997) where η is the hydraulic diffusivity. If we assume that $\eta = 1 \text{ m}^2/\text{s}$ and $r = 8 \text{ km}$, the time lag is estimated to be ~ 2 months. Although the assumed diffusivity is relatively large compared with typical values in crust (e.g., Li, 1985), this high diffusivity might be explained by assuming a well-connected fracture network as inferred from the contiguous alignment of seismicity in the studied area (Figure 1).

A plausible mechanism that can explain the increase in seismicity around spring is the unloading of surface mass due to melting snow. Heki (2003) reported that previous large inland earthquakes ($M \geq 7$) in snowy regions of the Japanese islands tend to occur more frequently in spring and summer than in fall and winter. Spring thaw causes unclamping of the fault planes below the snow cover, resulting in an increase in Coulomb failure stress by a few kilopascals at depth (Heki, 2003). In San-in district, the maximum snow depth reaches a few meters in mountainous regions during winter, and snow melting occurs around spring; the spring thaw is thus an important controlling factor of seismicity. Heki (2003) reported that smaller earthquakes do not show clear seasonal variations, but our results suggest that spring thaw affects both small and large earthquakes. This infers that monitoring the seasonal modulation of small earthquakes could lead to more accurate time-dependent forecasts of large earthquakes along the San-in seismic belt.

We suggest that a likely explanation of the seasonal modulation of seismicity is a combination of reduction in snow load in spring with increase in

rainfall amount in autumn. However, we cannot exclude that the other possible combination of different surface processes, such as pole-tides, thermoelastic and ocean loading (e.g., Ben-Zion & Allam, 2013; Dong et al., 2002; Johnson et al., 2017b), produces the semiannual modulation.

The present study implies that seasonal variation in seismicity could be potentially detected in an active tectonic zone characterized by large shear strain rate and/or high seismicity rate. Systematic investigation of seasonal variation in seismicity at different tectonic settings will allow us to deeply understand susceptibility of seismicity to surface processes.

5. Conclusions

We investigated seasonal modulation of crustal seismicity in San-in district using a new method that incorporates uncertainties derived from the declustering scheme itself. Background seismicity rates in the period 1980–2017 show significant semiannual variations, with increases in spring and autumn. Most large historic and modern earthquakes also occur in the same season, suggesting that the seismicity rate in San-in district has varied with the seasons for >150 years. An increase in crustal pore pressure due to precipitation in autumn and a decrease in snow load in spring might be plausible physical mechanisms to explain the observed variations.

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